

Data Management Project Proposal

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1 Project Typology

This project integrates two distinct Data Warehousing tracks:

1. **[DW]** Select a set of data sources, integrate them by means of ETL operations (you can either use an integration tool or write simple scripts to solve common ETL problems), define the DFM model for your data, define and populate the corresponding star schema or snowflake schema, by using a relational DB (e.g., Postgres). Perform OLAP queries.
Focus: ability to combine data from different datasets in order to produce a relevant unified source of analysis.
2. **[DW]** Identify a problem concerning data analysis and try to compare different approaches to address the proposed problem (e.g., carrying on the analysis via a relational DBMS and through a DW tool).
Focus: ability to compare different technologies

2 Datasets Description

To achieve the analysis, we integrate four data sources:

- **Dataset 1: Global EV Sales (2010–2024)**
Available on Kaggle, this dataset records annual electric vehicle sales and market trends across different countries. It tracks key metrics for Battery Electric Vehicles (BEVs), Fuel Cell Electric Vehicles (FCEVs), and Plug-in Hybrid Electric Vehicles (PHEVs).

Reference Link: <https://www.kaggle.com/datasets/patricklford/global-ev-sales-2010-2024>
- **Dataset 2: Our World in Data (OWID) CO₂ and Greenhouse Gas Emissions**
This repository includes comprehensive, regularly updated data on annual, per capita, cumulative, and consumption-based CO₂ emissions, alongside other greenhouse gases, population figures, and economic indicators.

Reference Link: <https://github.com/owid/co2-data.git>
- **Dataset 3: Our World in Data (OWID) Energy Dataset**
This repository supplies key metrics on primary energy consumption, consumption per capita, growth rates, global energy mix, electricity generation mix (renewables and fossil fuels), and other relevant metrics.

Reference Link: <https://github.com/owid/energy-data.git>
- **Dataset 4: World Bank Development Indicators**
This source includes Gross Domestic Product (GDP) metrics and other critical socioeconomic and development indicators across nations.

Reference Link: <https://databank.worldbank.org/source/world-development-indicators>

3 Project Overview

Through this project we are going to analyze environmental sustainability and electric mobility. The widespread adoption of electric vehicles is seen as the key to achieving zero transport emissions. However, an electric car is only truly “green” if the energy used to charge it comes from renewable sources.

The aim of this project is to determine whether the adoption of electric vehicles is actually leading to a real reduction in CO₂ emissions, or whether it is simply shifting the problem of pollution onto the electricity produced from fossil fuels.

The analysis will also focus on a financial perspective, by incorporating Gross Domestic Product (GDP) metrics to investigate how the transition to electric vehicles contrasts between wealthy nations and developing or poorer countries. We aim to explore whether higher economic capacity is correlated to the readiness to implement a clean energy production, or if poorer countries adopting EVs face to transitioning challenges in shifting their energy production towards sustainable sources due to capital constraints.